

Biodiversity Predictors

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1 Introduction

This project investigates marine biodiversity, focusing on how species richness responds to environmental and anthropogenic pressures. Marine ecosystems face mounting threats from climate change, overfishing, pollution, and habitat degradation. Recent evidence shows ocean warming is reshaping global biodiversity patterns, creating a more pronounced dip in species richness around the equator [Chaudhary et al., 2021]. Simultaneously, marine fish biodiversity in Arctic regions has doubled over three decades in response to rising sea temperatures [Gordó-Vilaseca et al., 2023], demonstrating that species redistributions are actively reshaping regional diversity patterns. While statistical learning methods are commonly used in ecology [Gordó-Vilaseca et al., 2023, Chaudhary et al., 2021, Braunisch, 2013, Hirzel et al., 2006], they have not been systematically compared on BioTIME’s marine data [Dornelas et al., 2025] to evaluate their effectiveness in modeling biodiversity-environment relationships.

2 Data and Problem Formulation

2.1 Dataset

2.2 Response Variable

2.3 Predictors

2.4 Problem Formulation

3 Methods

3.1 Regularized Linear Regression

3.1.1 Purpose

To model species richness while addressing potential multicollinearity and overfitting, we implemented regularized linear regression methods Ridge, Lasso, and Elastic Net. These approaches extend ordinary least squares (OLS) by adding

penalty terms to the loss function, which constrain coefficient estimates and improve generalization performance. All models were fit using the same set of predictors and a log-transformed response variable, log density.

3.1.2 Model Specifications

Ridge Ridge regression adds a squared penalty (L2) on the magnitude of coefficients, shrinking them toward zero but retaining all predictors in the model. This is particularly useful when predictors are highly correlated, as it stabilizes coefficient estimates without eliminating variables. Ridge also includes regularization hyperparameter alpha, controlling the strength of the penalty.

Lasso Lasso regression applies an absolute value penalty (L1), which can shrink some coefficients exactly to zero. This enables automatic feature selection, producing a sparse model that highlights the most influential environmental predictors of species richness. Ridge also includes hyperparameter alpha.

Elastic Net Elastic Net combines both L1 and L2 penalties, balancing the strengths of Ridge and Lasso. It is especially effective when predictors are highly correlated, as it can both stabilize coefficients and perform variable selection. This model includes alpha and mixing parameter l_1 ratio.

3.1.3 Model Evaluation

Model performance was evaluated using 10-fold cross-validation. Specifically, the data were randomly shuffled and partitioned into ten folds, and for each iteration the model was trained on nine folds and evaluated on the remaining fold. This process was repeated ten times so that each fold served once as the validation set. Predictive performance was summarized using the coefficient of determination (R^2), which measures the proportion of variance in the outcome explained by the model, and Root Mean Squared Error (RMSE).

- **Root Mean Squared Error (RMSE):** $\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$, measuring the average magnitude of the prediction error.

- R^2

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

In addition to predictive performance, model coefficients were examined to assess variable importance. Because all predictors were standardized within the modeling pipeline, coefficient magnitudes are directly comparable across variables.

3.2 Tree Based Methods

A decision tree algorithm was implemented from scratch in Python using weighted variance reduction as the split criterion. At each node, the algorithm searches all features and unique thresholds to find the combination that minimizes the weighted variance of the two resulting groups. We define this as

$$Weighted_Variance = \frac{n_{left}}{n_{total}}Var(left) + \frac{n_{right}}{n_{total}}Var(right).$$

This measures the impurity of each resulting group. Lower weighted variance indicated purer groups and more informative split. The tree grows recursively until either the maximum depth of 10 is reached, or fewer than 5 samples are remaining at a node, as traditionally used in class. This from scratch implementation was validated against sklearn’s DecisionTreeRegressor with identical parameters to confirm correctness.

The single decision tree was extended to a random forest ensemble using sklearn.ensemble.RandomForestRegressor. The random forest reduces variance over a single tree by averaging predictions across 100 trees, each trained on a bootstrap sample of the data with a random subset of features considered.

3.2.1 Model Evaluation

3.3 GAMs

3.3.1 GAM Purpose and Outcome

For outcomes specific to this study, we wanted to look at latitude-diversity gradients to determine whether richness peaks at tropics, poles or mid-latitudes non linearly. Secondly, we want to assess species richness across temperature ranges, to assess if we will see peaks or plateaus. Lastly we want to see the ocean depth effects to assess diminishing returns or flattening at extremes.

3.3.2 Model Selection and Justification

In order to answer these questions, we will apply a Generalized linear model (GLM): $y_i = \beta_0 + \sum_j \beta_j x_{ij}$, which assumes a linear relationship between predictors and response, with a Gaussian and an identity link: $g(\mu) = \mu$. Initially a negative binomial was considered but due to a numerical instability, Gaussian is a better practical approximation (see Appendix B.1). Commonly with species richness data we see that variance > mean which confirms that we would apply a negative binomial, else if mean = variance then a Poisson would be applied [Heit et al., 2024].

3.3.3 Smoothing and Penalization

To further capture the non-linear relationships we use Generalized Additive Models (GAMs): $y_i = \beta_0 + \sum_j f_j x_{ij} + \epsilon_i$, to replace the linear terms with penalized smooth terms $s(x)$; $s(\text{latitude})$, $s(\text{temperature})$, and $s(\text{depth})$, using

the same family and link. The penalized likelihood objective is in the form: $\sum (y_i - \hat{y}_i)^2 + \lambda \int (f''(t))^2 dt$. Then using generalized cross validation a lambda will be chosen to control the bias/variance trade off, so that the we likelihood gets penalized to control smoothness during model fitting. The effective degrees of freedom (edf) per smooth term quantifies the non-linearity of each relationship.

3.3.4 Evaluation Metrics

Model fit will be assessed using the following metrics:

- **Root Mean Squared Error (RMSE):** $\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$, measuring the average magnitude of the prediction error.
- **AIC:** $2k - 2 \ln(\hat{L})$, penalising model complexity by the number of parameters k against the log-likelihood $\hat{L}$. A lower value indicates a better trade-off.
- **Deviance Explained:** The proportion of null deviance reduced by the model, analogous to R^2 for GLMs and GAMs.
- **Residual Plots:** Will be used to assess heteroskedasticity, normality of errors, and systematic patterns in model assumptions.

4 Experimental Design

For all models we considered, our target variable was log-transformed species density:

$$species_density = \frac{species_richness}{AREA_SQ_KM}.$$

Raw species richness favors larger sampling areas naturally, which is not representative of the area's true diversity. This new feature makes sites comparable regardless of size. Further, the model would have been distorted by the heavy right skew of the species richness, so the log transformation produces a more symmetric distribution, better fit for regression, which we names *log_richness*.

$$log_richness = \log(species_density)$$

See appendix for the difference in the distributions, Figure (8).

4.1 Regularized Linear Regression

For our regression models, due to a high level of collinearity between many of the features, variance in coefficient estimates was highly inflated the models were very unstable. To address this we used the Variance Inflation Factor (VIF), using an iterative filtering procedure so that variables with VIF greater than

10 were sequentially removed. This resulted in the exclusion of the temperature proxy variable, resulting a final set of predictors with low collinearity and improved numerical stability.

The models were implemented with a pipeline framework that included the standardization of predictors using a z-score transformation followed by model fitting. This design ensured that scaling parameters were learned exclusively from the training data within each fold of cross-validation, preventing data leakage and ensuring valid performance estimates.

The dataset was split into training (80%) and testing (20%) subsets using a fixed random seed (42) to ensure reproducibility. Hyperparameters for each model were tuned using grid search with 5-fold cross-validation on the training set. For Ridge and Lasso regression, the regularization parameter (α) was selected over a logarithmically spaced grid, while Elastic Net additionally tuned the mixing parameter (l_1 ratio) to balance between L1 and L2 penalties.

4.2 Tree Based Methods

For tree based methods we use 14 environmental and geographic predictors that describe ocean depth, geographic location, climate, and sampling effort. We exclude *log_area* from the features since it is now the denominator of the target variable. See the appendix for the feature importance graph when we included *log_area*, Figure (9), it artificially inflated its feature importance to 0.60, masking other important indicators.

The data was split into 80/20 train and test splits with a fixed random seed (0) for reproducibility. The random forest was tested with 10 fold cross validation with RMSE and a primary metric and R^2 and AIC secondarily. The random forest was evaluated with cross validation using sklearn's `cross_val_score`.

Hyperparameters *max_depth* = 10 and *min_samples_leaf* = 5 were fixed across all tree variations to ensure fair comparison, matching defaults used in class exercises.

To check for robustness, two models were run. One with all 14 features including *sampling_intensity*, and one without to only address ecological predictors, removing potential methodological effects introduced by survey effort across studies.

4.3 GAMs

4.3.1 Data and Features

For this analysis there were three features: absolute latitude, temp proxy and mean depth, and predicted to log_richness as the target variable.

4.3.2 Train/test split

The dataset was split into a 80/20, Train and test set with a fixed random seed for reproducibility.

4.3.3 Smoothing and tuning

Using the Python package statsmodel and the model GLMGam handled the smoothing penalty selection internally by a generalized cross validation (GCV) when `gam.fit()` was applied to the model. B-splines with degree 3 and 6 degrees of freedom were used as the basis for each smooth term.

5 Results and Interpretation

5.1 Regularized Linear Regression

To evaluate the predictive performance of linear models while controlling for multicollinearity, we fit Ridge, Lasso, and Elastic Net regressions on the log-transformed species richness response.

Hyperparameters for each model were tuned using 5-fold cross-validation on the training set. Model performance was then evaluated on a held-out test set using R^2 and RMSE. All three regularized models performed very similarly, with low predictive power and high error. Ridge regression achieved an R^2 of 0.057 and RMSE of 1.141, while Lasso and Elastic Net produced similar results with R^2 values of approximately 0.056 and RMSE values of approximately 1.416. This suggests that the linear structure, even with added regularization, is insufficient to capture the complexity of the relationship between environmental predictors and species richness. This is often seen in ecological modeling.

To further assess model stability, 10-fold cross-validation was performed on the full dataset using the tuned pipelines. Results were highly consistent across folds and across models, with mean RMSE values of approximately 1.122 (± 0.049) and mean R^2 of approximately 0.062 (± 0.03). The near-identical performance across Ridge, Lasso, and Elastic Net indicates that neither coefficient shrinkage nor variable selection substantially improved predictive accuracy in this setting. Residual plots for the Lasso, Ridge, and Elastic Net models can be seen in Figure .

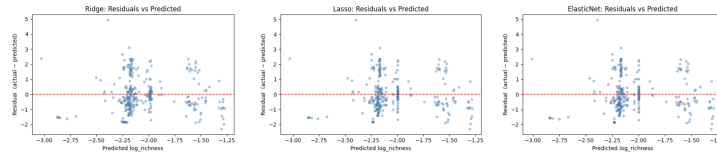


Figure 1: Regularized Regression Variable Importance

Despite weak predictive performance, coefficient estimates provide insight into variable importance. Across all models, latitude emerged as the strongest predictor with a consistently negative coefficient, indicating decreasing species richness at higher latitudes. The quadratic temperature term also showed a negative effect, suggesting non-linear thermal influences. Mean depth had a

Model	$RMSE$	R^2	AIC
From Scratch Decision Tree	0.733	0.585	-170.957
Sklearn Decision Tree	0.670	0.654	-229.679
Random Forest	0.638 ± 0.043	0.696 ± 0.032	-273.557
Random Forest (Eco)	0.657 ± 0.044	0.678 ± 0.034	-250.822

Table 1: Performance comparison of tree-based models predicting log marine species density. RMSE and R^2 for Random Forest models are 10-fold cross-validation estimates. *AIC is computed on the single test split for all models as AIC requires a single set of predictions.

positive association with richness, while precipitation proxy and sampling intensity had negligible or near-zero effects, with Lasso and Elastic Net shrinking these coefficients effectively to zero. The consistency of coefficient estimates across models reinforces the robustness of these relationships, even though their overall explanatory power is limited. The visual for variable importance across the three models can be seen in Figure.

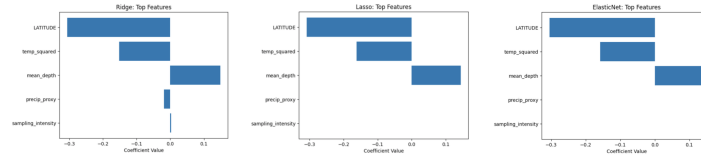


Figure 2: Regularized Regression Variable Importance

Overall, the poor performance of regularized linear models suggests that the relationship between environmental variables and marine species richness is inherently non-linear and may be better captured by more flexible modeling approaches such as tree-based methods or GAMs.

5.2 Tree Based Methods

The predictability performance in the two Decision Trees are very similar, showing that the from-scratch portion is correct. The performance gap can be accredited to sklearn’s more advanced optimized splitting techniques. The random forest outperforms each single tree by far, showing the benefit of averaging 100 diverse trees each using random groups of predictors.

5.2.1 Feature Importance

Once the sampling area was removed from the feature set, ecological predictors emerged as most important. Longitude was the strongest predictor of log species density at 32% importance, followed by mean ocean depth at 22% and latitude at 10%. The prominence of geographic coordinates reflects the ocean basin variation having distinct temperature regimes, currents, and histories to drive biodiversity differences. These importances were consistent with those found in

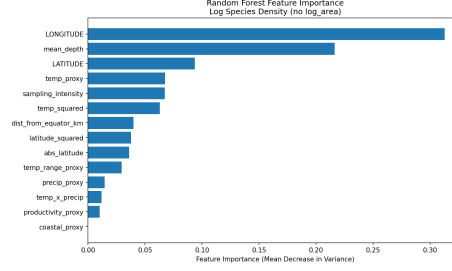


Figure 3: Random Forest feature importance rankings for predicting log marine species density. Importance is measured as mean decrease in variance across 100 trees. Log area was excluded from the feature set as it is mathematically embedded in the target variable denominator.

the regularized regression techniques. Coastal proximity contributed to lesser biodiversity, indicating that it does not add more information that what depth and location already provide.

5.2.2 Partial Dependence

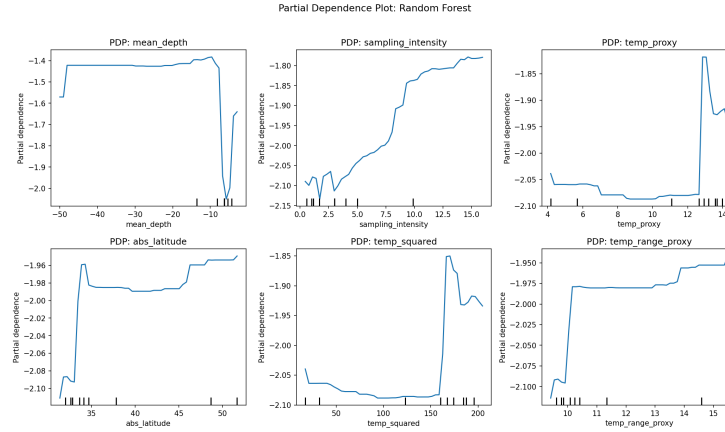


Figure 4: Partial dependence plots showing the marginal relationship between each predictor and log marine species density, holding all other variables constant. Plots are generated from the full random forest model.

Partial dependency plots (Figure 4) show the marginal relationships between each predictor and log species density. Ocean depth shows a flat relationship and then a very sharp drop at shallow sites, which could reflect human and natural disturbance in environments that are near the shore ([Henderson et al., 2020]). On the other hand, sampling intensity sharply peaks at low variables, and then again later on after a steady climb, which may depict that sites with few samples are rare spots with high diversity. Though this could also reflect differences in

methodology in the studies instead of ecological signals.

The longitude plot shows a sharp drop around 35 and a dip around -100, which separates the Indian and eastern Atlantic oceans from the Americas and western Pacific. The latitude plot shows highest density at low latitudes near the equator and then rise at higher latitudes. The temperature plot is mostly flat with large spikes at warmer climates. The temp proxy plot mirrors the latitude pattern as expected, given their mathematical relationship.

5.2.3 Robustness Check

To determine whether results were driven by sampling effort rather than ecological signal, *sampling_intensity* was removed from the feature set and the random forest was refitted. See Table 1.

The R^2 drops by very slightly, suggesting that ecological variables still explain the majority of the variance in log species density, confirming that the environmental impact is not merely an artifact of survey effort across studies.

Without sampling intensity, *mean_depth* rises from 22% to 35% importance and temperature squared and proxy are following with 13% each. These features emerge to be the primary drivers of marine species diversity. The lower AIC of the ecological model suggests it achieves a better balance of fit despite the slight performance drop.

5.2.4 Error Analysis

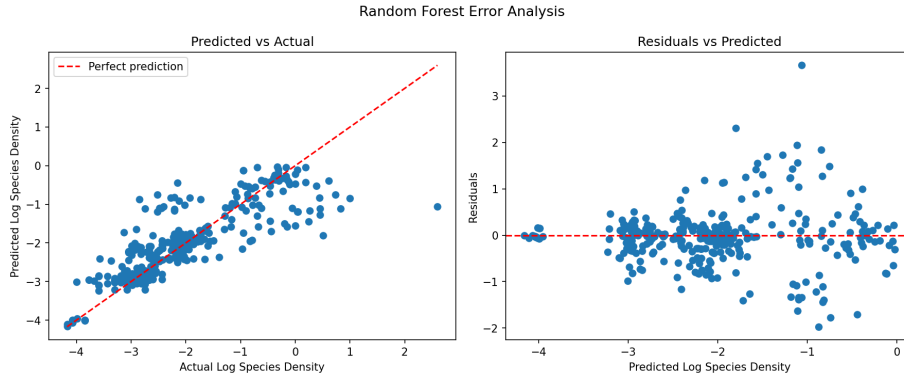


Figure 5: Random Forest error analysis for log marine species density. Left: predicted versus actual values with perfect prediction line shown in red. Right: residuals versus predicted values. Residuals are approximately centered around zero with mild heteroscedasticity at higher predicted values.

The first plot in Figure 5 shows points closely following the diagonal across the range, indicating ideal function of the trends. It performs best in the denser sections, and much worse at the high extremes above log 0, where we see constant under prediction. This is expected, as Random Forests tend to pull extreme

predictions to the mean. In the second plot we see residuals approximately centered around zero with no directional bias. The variance is noticeably larger at higher predicted values, which may reflect genuine ecological complexity at high densities that span a larger range of environments.

5.3 GAMs

5.3.1 Model Performance

Metric	Value
RMSE	1.111
AIC	3996.334
Deviance Explained	0.062
R^2	0.104

Table 2

The RMSE result tells us that the predictions of the model are off by around 1.1 log-richness units on average. The model explains 10% of variance in log species richness with a deviance explained of 0.062. Which is reasonable for having three predictors on our biodiversity dataset [Xue et al., 2018].

5.3.2 Partial Effect Plots

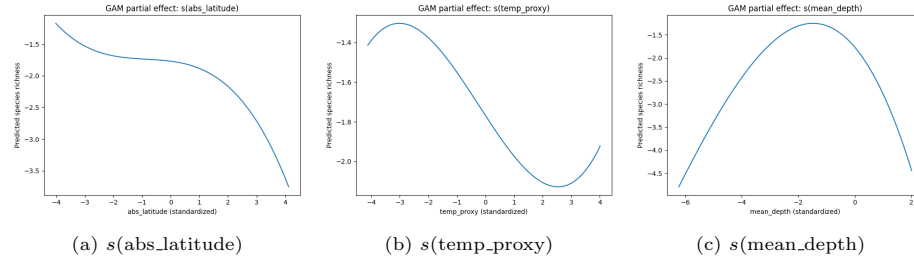


Figure 6: Partial effect plots for the three GAM smooth terms.

$s(\text{abs_latitude})$

There is a monotonic decline, where richness decreases as latitude increases with the steepest drop in the polar regions (high altitudes). This aligns with the latitude-diversity gradient as mentioned by [Chaudhary et al., 2021].

$s(\text{temp_proxy})$

There is a peak in richness at cold/cooler temperatures and declines through warm temperatures, with a recovery at the hottest extreme. The recovery at

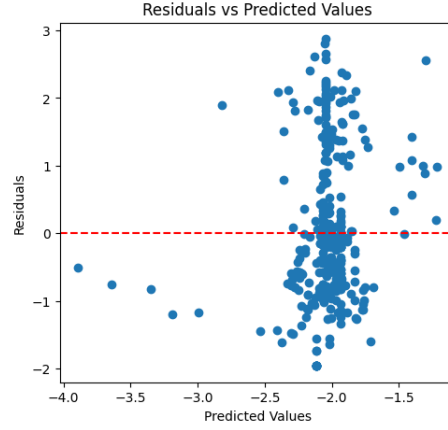


Figure 7: Enter Caption

the hottest extreme could be due to sparse data at temperature extremes rather than an answer to the richness in in that area.

$s(\text{mean_depth})$

There is a high peak of richness at mid depths, with sharp drop offs at the extreme shallow and deep ends. This confirms the diminishing returns hypothesis that very deep sites in the ocean have lower species richness [Henderson et al., 2020].

Across the three smooth terms depth shows the strongest non-linear structure with the largest range in the predicted value (log-richness), which concludes that it is the most influential predictor in shaping species richness patterns. There is no single predictor dominating in isolation, as our results suggest that marine species richness in temperature environments is driven by geographic position(latitude), thermal conditions(temp) and bathymetric context(depth)

5.3.3 Residual Analysis

The plot shows residuals are clustered around predicted values, within the range of log-richness in the dataset. We can see that variance is not constant, and can say that this is heteroskedasticity and violated the Gaussian assumption of constant variance. The sparse predictions suggest the model struggles with low-richness sites.

6 Conclusions, Limitations, Future Work

7 References

References

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8 AI Statement

A Appendix

A.1 From-Scratch Implementation

```
1 # lower weighted variance = purer groups = better split
2 def weighted_variance(left, right):
3     n_total = len(left) + len(right)
4     weight_left = len(left) / n_total
5     weight_right = len(right) / n_total
6     return (weight_left * np.var(left)) + (weight_right * np.var(
7         right))
8
9 # searches all features and thresholds to find the split minimizing
10 # weighted variance
11 def best_split(X, y):
12     best_feature = None
13     best_threshold = None
14     best_score = float('inf')
15
16     if len(y) < 2:
17         return None, None, float('inf')
18
19     n_features = X.shape[1]
20
21     for feature_idx in range(n_features):
22         thresholds = np.unique(X[:, feature_idx])
23
24         for threshold in thresholds:
25             left_mask = X[:, feature_idx] <= threshold
26             right_mask = ~left_mask
27
28             left_y = y[left_mask]
29             right_y = y[right_mask]
30
31             if len(left_y) == 0 or len(right_y) == 0:
32                 continue
33
34             score = weighted_variance(left_y, right_y)
35
36             if score < best_score:
37                 best_feature = feature_idx
38                 best_threshold = threshold
39                 best_score = score
40
41     return best_feature, best_threshold, best_score
42
43 # recursively builds the tree and stops at max_depth or
44 # min_samples_leaf
45 def build_tree(X, y, depth=0, max_depth=10, min_samples_leaf=5):
46     # Stopping conditions to return a leaf node
47     if depth >= max_depth or len(y) < min_samples_leaf:
48         return {'value': np.mean(y)}
```

```

48     feature, threshold, score = best_split(X, y)
49
50     if feature is None:
51         return {'value': np.mean(y)}
52
53     left_mask = X[:, feature] <= threshold
54     right_mask = ~left_mask
55
56     left_subtree = build_tree(X[left_mask], y[left_mask], depth +
57                               1, max_depth, min_samples_leaf)
58     right_subtree = build_tree(X[right_mask], y[right_mask], depth
59                               + 1, max_depth, min_samples_leaf)
60
61     return {'feature': feature, 'threshold': threshold, 'left':
62           left_subtree, 'right': right_subtree}
63
64 # makes tree for one site
65 def predict_one(node, row):
66     if 'value' in node:
67         return node['value']
68     if row[node['feature']] <= node['threshold']:
69         return predict_one(node['left'], row)
70     else:
71         return predict_one(node['right'], row)
72
73 # predicts for all sites
74 def predict(node, X):
75     return np.array([predict_one(node, row) for row in X])

```

A.2 Additional Figures

B Additional Figures

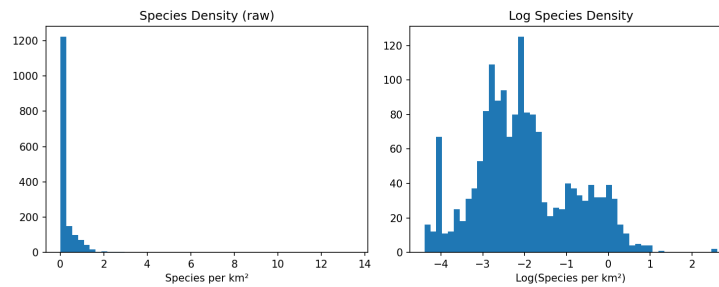


Figure 8: Distribution of species density before and after log transformation. Raw species density (left) is heavily right-skewed with most sites clustered near zero and a small number of extreme outliers up to 14 species per km². Log-transformed density (right) produces a approximately symmetric distribution suitable for regression.

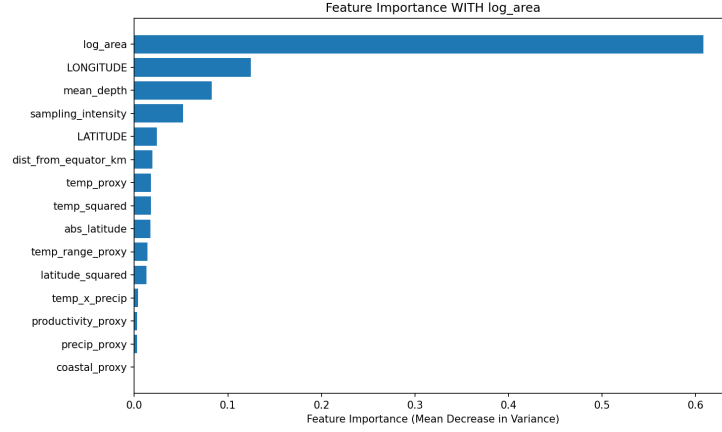


Figure 9: Random Forest feature importance when log area is included in the feature set. Log area dominates at 60% importance, artificially suppressing all ecological predictors. This circular relationship arises because area is embedded in the denominator of the target variable (log species density), motivating its exclusion from the main analysis.

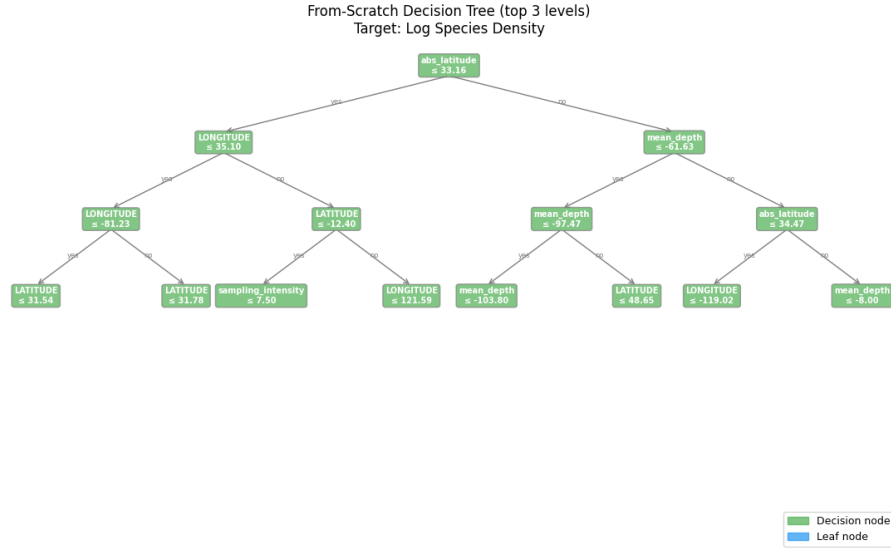


Figure 10: From-scratch decision tree structure showing the top three levels. Green nodes are internal decision nodes showing the feature and threshold used for splitting. The tree first partitions on absolute latitude at 33.16°, corresponding to the subtropical boundary, then subdivides on longitude and mean depth at deeper levels.

B.1 Model Distribution Selection

we can see that the variance is much larger than the mean, which would guide us to have used negative binomial as we have confirmed overdispersion. How-

Statistic	Value
Mean	-2.047
Variance	1.352

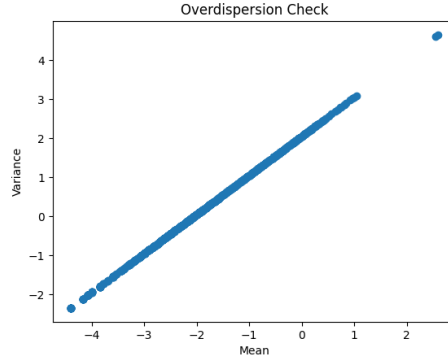


Figure 11: Overdispersion check

ever the Python package statsmodels, and its function GLMGam has a limitation where negative binomial fails to converge with the backfitting algorithm (LinAlgError) used internally [Statsmodels Developers, 2024]. Thus Gaussian was used as a practical alternative. Gaussian is a practical approximation with our log-transformed species richness as the target, as it helps address the count nature of the response.